QQ-Z-100A

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July 6, 1964

SUPERSEDING
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April 11, 1958

FEDERAL SPECIFICATION

ZINC ALLOY SHEET AND STRIP

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two grades of zinc-alloy products (sheet and strip) having slit or sheared edges (see 6.1), for the procurement of rain goods, such as roofing, flashing, and miscellaneous sheet work.

1.2 Classification.

1.2.1 *Grades.* Zinc-alloy sheets and strips shall be of the following grades, as specified (see 6.2):

Grade 1 — Copper hardened alloy.

Grade 2 — Copper-titanium alloy (creep resistant alloy).

1.2.2 *Forms.* Zinc-alloy shall be furnished as pack rolled sheet or as strip, either in coils or flat lengths (see 6.2).

1.2.3 *Temper.* Grade 1 zinc alloy shall be furnished hot rolled or cold rolled, as specified (see 6.2). Grade 2 zinc alloy shall be furnished hot rolled, or rolled and heat treated to develop creep resistance, as specified (see 6.2).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids form a part of this specification to the extent specified herein.

Federal Specifications:

PPP–B–601 — Boxes, Wood, Cleated Plywood.
PPP–B–621 — Boxes, Wood, Nailed and Lock-corner.
PPP–C–650 — Crates, Wood, Open and Covered.

Federal Standards:

Fed. Std. No. 102 — Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 — Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151 — Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are avail-

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able without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM) Publications:

E 3 — Preparation of Metallographic Specimens.

E 79 — Estimating the Average Grain Size of Wrought Copper and Copper-Base Alloys.

E 80 — Rec. Practice for Dilatometric Analysis of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Chemical composition. The zinc-alloy sheet and strip shall meet the chemical composition requirements of table I.

TABLE I.—*Composition of zinc alloy*

	Grade 1	Grade 2
Copper, percent	0.2 to 1.2	0.5 to 0.8
Lead, percent, max.	0.25	0.20
Iron, percent, max.	.015	.015
Chromium, percent, max.	.02	.02
Manganese, percent, max.	.01	.01
Cadmium, percent, max.	.07	.01
Titanium, percent	.....	.08 to 0.16
Zinc	Remainder	Remainder

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3.2 Mechanical properties.

3.2.1 *Tensile properties.* Zinc Alloy sheet and strip shall meet the tensile requirements shown in table II, at ¼ inch per minute free-running crosshead speed.

TABLE II.—*Tensile strength requirements*

Item	Tensile strength, p.s.i. minimum	
	Grade 1	*Grade 2*
Sheet	22,000	24,000
Strip	18,000	24,000

3.2.1.1 *Elongation in 2 inch gage length.* Zinc-alloy sheet and strip shall meet the minimum elongation shown in table III.

TABLE III.—*Elongation in 2-inch gage length*

Item	Percent elongation, minimum	
	Grade 1	*Grade 2*
Sheet	5	10
Strip	15	10

3.2.2 *Hardness.* Zinc alloy sheet and strip shall meet the minimum hardness shown in table IV.

TABLE IV.—*Rockwell hardness requirements, 15T scale for 15 seconds, minimum*

Item	Hardness	
	Grade 1	*Grade 2*
Sheet	55	55
Strip	45	55

3.2.3 *Bend properties.* Bend test specimens taken from zinc-alloy sheet and strip shall withstand, without cracking, the bend test specified in 4.5.3.

3.3 Coefficient of thermal expansion. When tested in accordance with 4.5.5, the coefficient of thermal expansion for grade 2 sheet and strips shall be:

With grain — 0.000015 in./in./° F. maximum.

Across grain — 0.000012 in./in./° F. maximum.

3.4 Resistance to grain growth. After heat treatment for 4 hours at 650° F., the average grain size of grade 2 sheet and strip, when measured in polarized light, shall not exceed 0.0005 inch.

3.5 Dimensions. The dimensions of the zinc-alloy sheet or strip shall be specified in the contract or order (see 6.2).

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3.6 Dimensional tolerances.

3.6.1 *Thickness.* Zinc alloy sheets and strips shall not vary from the ordered thickness by more than the amount shown in table V.

TABLE V.—*Permissible variation in thickness of zinc-alloy sheet and strip*

Ordered thickness	Thickness tolerance, plus and minus, for the ordered width [1]						
	Up to 8 incl.	Over 8 to 12 incl.	Over 12 to 14 incl.	Over 14 to 20 incl.	Over 28 to 28 incl.	Over 28 to 36 incl.	Over 36 to 60 incl.
.017 to .021 incl.	0.0013	0.0018	0.0018	0.0020	0.0030	0.0035	0.0040
over .021 to .026 incl.	.0015	.0020	.0020	.0025	.0030	.0035	.0040
over .026 to .037 incl.	.0020	.0020	.0020	.0025	.0035	.0040	.0050
over .037 to .050 incl.	.0020	.0025	.0025	.0030	.0040	.0050	.0060
over .050 to .075 incl.	.0025	.0030	.0030	.0035	.0045	.0055	.0065
over .075 to .100 incl.	.0030	.0035	.0035	.0040	.0050	.0060	.0070
over .100 to .125 incl.	.0040	.0045	.0045	.0050	.0060	.0065	.0075

[1] All dimensions are in inches.

3.6.2 *Width.* Zinc alloy sheet and strips shall not vary from the ordered width by more than the amounts shown in tables VI and VII.

TABLE VI.—*Permissible variation in width for flat products, square, sheared metal*

Size ordered	Width tolerances, [1] plus or minus (inch)		
Ordered width, inches	Ordered thickness up to 1/16 inch incl.	Ordered thickness over 1/16 inch to 1/8 inch incl.	Ordered thickness over 1/8 inch
Up to 20, incl	1/32	3/64	1/16
Over 20 to 36	3/64	3/64	1/16
Over 36 to 60	1/16	1/16	1/16

[1] If tolerances are desired all plus or all minus, double the values given.

TABLE VII.—*Permissible variation in width for flat products, slit metal*

	Width tolerance, [1] plus or minus (inch)		
Ordered width, inches	For thickness .017 to .032 inch incl.	For thickness .032 to 0.125 inch incl.	For thickness over 0.125 inch
Up to 2	0.0100	0.0200	0.0300
Over 2 to 8 incl.	.0150	.0250	.0400
Over 8 to 20 incl.	.0200	.0300	.0500

[1] If tolerances are desired all plus or minus, double the values given.

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3.6.3 *Length.*

3.6.3.1 *Metal with slit or sheared edges.*

3.6.3.1.1 Zinc alloy with slit or sheared edges shall not vary from the ordered length by more than the amounts shown in table VIII.

TABLE VIII.—*Permissible variation in length for flat products with slit or sheared edges*

Ordered length	Permissible variation in length, plus only
	Inch
Specified lengths up to 10 feet incl.	¼
Specified length over 10 feet up to 12 feet incl.	½

3.6.3.2 *Metal with edges and ends sheared.*

3.6.3.2.1 Flat products with edges and ends sheared shall not vary from the ordered length by more than the amounts shown in table VIII.

3.6.4 *Straightness.*

3.6.4.1 *Slit metal.* The edgewise curvature (depth of arc) in a 6-foot portion of the total length shall not exceed the amounts shown in table IX.

TABLE IX.—*Permissible variation in straightness for slit metal*

Ordered width	Tolerances		
	As slit edges		As slit and straightened
	Shipped in rolls	Shipped flat	Shipped flat or in rolls
	Inches	*Inch*	*Inch*
Over ¼ to ⅜ incl.	1½	¾	¼
Over ⅜ to ½ incl.	1	½	¼
Over ½ to 1 incl.	⅝	⅜	¼
Over 1 to 2 incl.	⅜	5/16	¼
Over 2 to 4 incl.	5/16	¼	¼
Over 4	¼	¼	¼

3.6.4.2 *Square-sheared metal.* The edgewise curvature (depth of arc) in any 6-foot portion of the total length shall not exceed the amounts shown in table X.

TABLE X.—*Permissible variation in straightness for square sheared metal* [1]

Ordered thickness	Tolerances	
	For widths up to 10 inches incl.	For widths over 10 inches
Inch	*Inch*	*Inch*
Up to ⅛ incl.	⅛	1/16
Over ⅛ to 3/16	⅛	1/16
Over 3/16	⅛	1/16

[1] Not applicable to metal over 10 feet in length.

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3.6.4.3 *Stretcher leveling.* When furnished, sheets shall be stretcher leveled.

3.7 Identification marking.

3.7.1 *Sheet and strip.* Unless otherwise specified (see 6.2), marking of sheet and strip shall be in accordance with the manufacturer's standard practice. Additional marking requirements shall be specified in the contract or order.

3.8 Workmanship. The zinc-alloy sheet and strip shall be uniform in quality, sound, commercially clean, flat, and smooth, and shall be commercially free from injurious defects including foreign material, pipes, slivers, laps, cracks, seams, scales, burrs, buckles, damaged ends, damaged corners, and damaged edges.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified (see 6.2), a lot shall consist of not more than 10,000 pounds of material of the same grade, form, temper, and size.

4.3 Sampling.

4.3.1 *Sampling for examination.* Sampling for examination shall be in accordance with MIL–STD–105, inspection level II. Acceptance and rejection shall be based on an acceptance quality level (AQL) of 1.5.

4.3.2 *Sampling for test.* Sampling for tension, hardness, and bend tests shall be in accordance with MIL–STD–105, inspection level S–2. Acceptance and rejection shall be based on an AQL of 1.0.

4.3.3 *Sampling for chemical analysis.* Samples shall be randomly selected from each lot in accordance with table XI.

TABLE XI.—*Sampling for chemical analysis*

Pounds of material in lot	Number of samples
500 and under	1
501 to 2000	2
2001 to 5000	3
5001 to 10,000	4

4.4 Examination. Each of the sample pieces selected in accordance with 4.3.1 shall be visually and dimensionally examined to determine compliance with 3.5 to 3.7, inclusive.

4.4.1 *Straightness.* Straightness shall be determined by placing the piece on a level surface so that the arc or departure from straightness is horizontal. The maximum depth of arc shall be measured to the nearest $\frac{1}{32}$ inch by means of a straight edge and steel scale.

4.5 Tests.

4.5.1 *Tension test.* A tension test specimen shall be machined from each of two sample pieces selected in accordance with 4.3.2, to the form and dimensions of type F2, method 211, of Fed. Test Method Std. No. 151. The longitudinal axis of the specimens shall be parallel to the direction of rolling.

4.5.1.1 *Test method.* All tension tests shall be conducted in accordance with method 211 of Fed. Test Method Std. No. 151. The speed of testing shall be $\frac{1}{4}$ inch per minute.

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4.5.1.2 *Elongation.* Elongation shall be measured in accordance with method 211 of Fed. Test Method Std. No. 151.

4.5.2 *Hardness test.* Each of the sample pieces selected in accordance with 4.3.2 shall be tested for hardness in accordance with method 243 of Fed. Test Method Std. No. 151.

4.5.3 *Bend test.* Bend test specimens shall be prepared from each of the sample pieces selected in accordance with 4.3.2. Specimens shall meet the applicable requirements of method 231 of Fed. Test Method Std. No. 151, and shall be bent either parallel with or transversely to the direction of final rolling, at room temperature through 180° F. around a mandrel having a diameter equal to N times the nominal thickness of the sheet. The values for N are given in table XII. When zinc-alloy sheet and strip are ordered in thicknesses greater than .050 inch, the bend factor (N) shall be specified in the contract or order (see 6.2). Specimens shall be bent by either pressure or blows. Specimens subjected to the bend test shall be examined for compliance with the requirements.

TABLE XII.—*Bend factor*

Type	Nominal thickness	Bend factor
	Inch	*N*
Sheet & Strip	0.017 to 0.020	0
	.021 to .032	1
	.032 to .040	2
	.040 to .050	3

4.5.4 *Chemical analysis.* Unless otherwise specified (see 6.2), samples selected in accordance with 4.3.3 shall be analyzed by method 111 or 112 of Fed. Test Method Std. No. 151.

4.5.5 *Coefficient of thermal expansion.* Unless otherwise specified (see 6.2), the test for coefficient of thermal expansion shall be in accordance with ASTM E 80 and shall meet the requirements of 3.3.

4.5.6 *Grain size.* Grain size determination shall be in accordance with ASTM E 79, except that grain size shall be determined in inches. The specimens shall be prepared in accordance with 4.5.6.1.

4.5.6.1 Unless otherwise specified (see 6.2), preparation of the metallographic specimens shall be as follows. The specimens shall be prepared in accordance with ASTM E 3. Specimens should be etched between the wet polishing stages to minimize distortion. The final polishing and etching steps should be alternately repeated until no change in structure is observed. The final etching should be five seconds in the following solution:

20 grams pure CrO_3

1.5 grams Na_2SO_4 or 3.4 grams $Na_2SO_4 \cdot 10H_2O$.

100 ml. of water

Etching is followed by a rinse in a solution containing 20 grams pure CrO_3 in 100 ml. of water to avoid staining, then rinsed in water and dried.

4.6 Examination of preparation for delivery. The packing and marking of the sheets and strips shall be examined to determine compliance with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of various levels of packing for civil agency procurements.)

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* Sheets and strips shall be packed in containers conforming to the requirements of PPP–B–601 (overseas type);

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PPP–B–621 (class 2); and PPP–C–650. Flat steel strapping shall be type 1, class B. Boxes exceeding 250 pounds gross weight shall be modified by the addition of skids in accordance with the box specification.

5.1.2 *Level B.* Sheets and strips shall be packed as specified for level A except that containers shall be domestic type and class, as applicable.

5.1.3 *Level C.* Sheets and strips shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. This specification covers two grades of zinc-alloy sheet and strip for general purposes, such as, the manufacture of gutters, conductor pipe, flashings, and general sheet metal work required in building construction.

6.1.1 The superior qualities of grade 2 alloy makes it suitable for all sheet metal applications indicated in 1.1 and 6.1.

6.1.2 Grade 2 alloy should be used for all applications where creep resistance and low-thermal expansion are required. Grade 1 alloy should not be used for applications where these characteristics are critical.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specifiy the following:

(*a*) Title, number, and date of this specification.

(*b*) Grade of alloy (see 1.2.1).

(*c*) Form required (see 1.2.2).

(*d*) Temper required (see 1.2.3).

(*e*) Dimensions (see 3.5).

(*f*) Marking required (see 3.7.1).

(*g*) When a lot shall be other than specified (see 4.2).

(*h*) Bend factor, if required (see 4.5.3).

(*i*) Chemical analysis, if different (see 4.5.4).

(*j*) Thermal expansion test, if different (see 4.5.5).

(*k*) Preparation of specimens, if different (see 4.5.6.1).

(*l*) Selection of applicable level of packing required (see 5.1).

6.3 Zinc-alloy should be formed at temperatures above 60° F. Under no circumstances should any forming be done at temperatures under 50° F. Pittsburgh lock seams can be readily formed; however, care should be taken to avoid burrs and sharp angle bends. A minimum radius of 1/16 inch is recommended for all bends.

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Zinc-alloy, sheet or strip, not otherwise indexed by name. Carload minimum weight 36,000 pounds.

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Motor:

Zinc-alloy, sheet or strip, not otherwise indexed. Truckload minimum weight 36,000 pounds, subject to rule 115, National Motor Freight Classification.

CUSTODIANS:

Army—MO
Navy—Docks
Air Force—MOA

Other interests:

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